python

WsCube Tech
Upskilling Bharat

Super Market Billing System

PYTHON-BASED RETAIL BILLING
AUTOMATION SYSTEM

PRESENTED BY: DIVYA SHARMA

Introduction

Developed a Python based supermarket billing system to automate the checkout process.

Calculates item prices, applies discounts, and generates accurate bills instantly.

Helps reduce manual errors and speeds up customer transactions.

Demonstrates the use of automation to improve efficiency in retail operations.

Project Setup & Fundamental Logic: Supermarket Billing System

Project Workspace: 01

Set up a dedicated folder (supermarket_billing_system) and use .ipynb files for interactive, cell-based coding.

Variable Management: 02

Use Strings for item names, Integers for quantities, and Floats for decimal prices to ensure data accuracy.

Data Storage: 03

Implement Dictionaries ({Item: Price}) to create a searchable database of store inventory and pricing.

Automated Logic: 04

Use arithmetic operators for totals (Price * Qty) and f-strings to generate clean, formatted digital receipts.

Key Functional Requirements

Multi-Item Entry

Use Loops to handle continuous input until the billing process is finished.

Dynamic Discounts

Use Conditional Statements (if-else) to apply discounts based on item category or total bill amount.

Itemized Reporting

Generate a receipt displaying names, quantities, individual totals, and final amounts.

supermarket_billing_system

supermarket.ipynb

Billing Input

Supermarket Billing System

This project calculates the total bill for multiple items, applies a discount, and generates a simple receipt using Python.

```python
from datetime import datetime

print("="*50)
print("        WELCOME TO SMART BILLING SYSTEM")
print("="*50)

cart = []
subtotal = 0

while True:
    name = input("\nEnter item name: ")
    qty = int(input("Enter quantity: "))
    price = float(input("Enter price per unit: "))
    category = input("Enter category (grocery/beauty/electronics/other): ").lower()

    total = qty * price

    # Category discount
    discount = 0
    if category == "beauty":
        discount = total * 0.05
    elif category == "grocery":
        discount = total * 0.02

    final_price = total - discount
    subtotal += final_price

    cart.append([name, qty, price, total, discount, final_price])

    more = input("Do you want to add another item? (yes/no): ")
    if more.lower() != "yes":
        break
```

Receipt Generation

```python
# -------- PRINT RECEIPT -------- #

print("\n\n")
print("="*60)
print("                    SMART MART")
print("              GST INVOICE / RECEIPT")
print("="*60)

now = datetime.now()
print("Date:", now.strftime("%d-%m-%Y"))
print("Time:", now.strftime("%H:%M:%S"))
print("-"*60)

print(f"{'Item':18}{'Qty':<6}{'Price':<10}{'Total':<10}{'Disc':<8}{'Final'}")
print("-"*60)

for item in cart:
    print(f"{item[0]:18}{item[1]:<6}{'₹'+format(item[2],'.2f'):<10}"
          f"{'₹'+format(item[3],'.2f'):<10}{'₹'+format(item[4],'.2f'):<8}"
          f"₹{item[5]:.2f}")

print("-"*60)

# Bill level discount
bill_discount = 0
if subtotal > 2000:
    bill_discount = subtotal * 0.10

final_amount = subtotal - bill_discount

print(f"{'Subtotal:':40} ₹{subtotal:.2f}")
print(f"{'Bill Discount:':40} -₹{bill_discount:.2f}")
print("="*60)
print(f"{'FINAL AMOUNT TO PAY:':40} ₹{final_amount:.2f}")
print("="*60)

print("        Thank you for shopping with us!")
print("              Visit Again 😊")
print("="*60)
```

Smart Billing System

```
=====================================================
        WELCOME TO SMART BILLING SYSTEM
=====================================================
```

```
==============================================================
                  SMART MART
            GST INVOICE / RECEIPT
==============================================================
Date: 14-02-2026
Time: 01:02:17
--------------------------------------------------------------
Item              Qty   Price     Total     Disc    Final
--------------------------------------------------------------
Rice              5     ₹80.00    ₹400.00   ₹8.00   ₹392.00
Shampoo           2     ₹250.00   ₹500.00   ₹25.00  ₹475.00
Earphones         1     ₹1200.00  ₹1200.00  ₹0.00   ₹1200.00
cooking oil       3     ₹700.00   ₹2100.00  ₹42.00  ₹2058.00
face wash         1     ₹130.00   ₹130.00   ₹6.50   ₹123.50
--------------------------------------------------------------
Subtotal:                                   ₹4248.50
Bill Discount:                              -₹424.85
==============================================================
FINAL AMOUNT TO PAY:                        ₹3823.65
==============================================================
        Thank you for shopping with us!
              Visit Again 😊
==============================================================
```

Billing Input Logic – Logic and Implementation

01

Importing datetime

Reason: To fetch the current system date and time, ensuring every receipt is unique and time-stamped for record-keeping.

02

Initializing Variables (cart and subtotal):

Reason: Creates a "storage box" (cart) for items and a "counter" (subtotal) to track the running cost.

03

The while True Loop:

Reason: This creates an interactive "entry mode" that stays open so the cashier can scan an unlimited number of items without restarting the program.

04

Input & Type Casting:

Reason: input() always treats data as text; using int() and float() converts that text into numbers so the computer can perform multiplication and subtraction.

05

Base Total Calculation

Reason: Multiplies qty * price to find the cost of the current item before any discounts are applied.

06

Category-Based if-elif Logic

Reason: Automatically identifies "Beauty" or "Grocery" items to apply specific store promotions (5% or 2% off).

07

Updating the Subtotal

Reason: Adds the final_price of the current item to the grand total (subtotal += final_price) as you go.

08

Loop Control (Break Condition)

Reason: Asks if there are more items; if the user says anything other than "yes," it stops the input phase and moves to the bill.

Receipt Generation – Logic and Implementation

Print Receipt

01 Section Break & Headers

Reason: Uses print and \n to clear space and create a bold title ("SMART MART"), making the receipt look professional and easy to read.

02 Fetching Current Time

Reason: Calls datetime.now() to capture the exact second the bill is generated, which is necessary for business records and customer disputes.

03 Time Formatting (strftime)

Reason: Calls datetime.now() to capture the exact second the bill is generated, which is necessary for business records and customer disputes.

04 Table Header Alignment

Reason: Uses F-strings with fixed widths (like :18) to create a stable header row so that "Item," "Qty," and "Price" columns never shift.

05 The for Loop Iteration

Reason: Steps through the cart list one by one to print every item the customer bought; without this, only the last item would show up.

06 Currency & Decimal Formatting

Reason: Uses ₹ and .2f to ensure all money values show exactly two decimal places (e.g., ₹10.00), matching standard banking formats.

07 Bulk Discount Calculation

Reason: Applies a final if check to see if the total is over ₹2000, rewarding big spenders with an extra 10% discount before the final total.

08 Grand Total & Footer

Reason: Calculates final_amount and prints it between double-line borders (=) to clearly show the customer exactly what they need to pay.

python

WsCube Tech
Upskilling Bharat

Thank you

www.linkedin.com/in/divya9916

github.com/Divya9916